



YSI 6820 and 6920 VZ Sondes

With 1 or 2 optical ports and new sensor options



The YSI 6820 VZ-2 and 6920 VZ-2 Sondes

Measure multiple parameters simultaneously including:

Temperature
Conductivity
Specific Conductance
Salinity
Resistivity
TDS
pH
ORP
Depth or Level
Nitrate, Ammonium or Chloride
Rapid Pulse™ DO (VZ-1 only)

And 1 or 2 of the following optical sensors:

ROX™ Optical DO
Turbidity
Chlorophyll
Blue-Green Algae
(Phycocyanin or Phycoerythrin)
Rhodamine

- Two bulkhead versions available:
 - The 6820/6920 VZ-1 has one optical port, conductivity/temperature port, Rapid Pulse™ DO port, pH/ORP port, and three ISE ports
 - The 6820/6920 VZ-2 has two optical ports, conductivity/temperature port, pH/ORP port, and one ISE port
- Self-cleaning optical sensors with improved wiping
- Field-replaceable sensors
- 6920 VZ has a built-in battery compartment for long-term *in situ* monitoring

Take Advantage of YSI's New Optical Sensors

In addition to turbidity, chlorophyll, and rhodamine, YSI now offers these optical sensors:

ROX Reliable Optical Dissolved Oxygen

The ROX sensor uses lifetime luminescence detection technology to offer the most reliable oxygen sensor with the lowest possible maintenance effort. Experience significantly less membrane maintenance while obtaining excellent accuracy, sensitivity, and range.



Blue-Green Algae (BGA)

YSI's fluorescence-based blue-green algae sensors will allow you to monitor blue-green algae populations where their presence is a concern. Whether providing an early warning to an algal bloom, tracking taste and odor-causing species in drinking water supplies, or conducting ecosystem research; YSI BGA sensors will provide sensitive and reliable *in situ* data.

6820 and 6920 Upgrades Available

YSI is committed to offering our customers reliable and cost-effective water monitoring solutions. To this end, we are offering VZ-2 Upgrades for existing 6820/6920s. Upgrades will be available from YSI Authorized Service Centers and will include the new 6820/6920 VZ-2 bulkhead, an Optical Dissolved Oxygen Sensor, and firmware/software upgrades. In addition, the sonde will be fully tested and calibrated by an experienced YSI service technician.

Pure
Data for a
Healthy
Planet.®

*Compact sondes for
field sampling and data
collection platforms*



To order, or for more info,
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ISO 9001
ISO 14001

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*Sensors with listed with the ETV logo were submitted to the ETV
program on the YSI 6600EDS. Information on the performance
characteristics of YSI water quality sensors can be found at www.
epa.gov/etv, or call YSI at 800.897.4151 for the ETV verification
report. Use of the ETV name or logo does not imply approval
or certification of this product nor does it make any explicit or
implied warranties or guarantees as to product performance.

YSI incorporated
Who's Minding
the Planet?

YSI 6820 V2 & 6920 V2 Sensor Specifications

	Range	Resolution	Accuracy
ROX™ Optical Dissolved Oxygen* % Saturation	0 to 500%	0.1%	0 to 200%: ±1% of reading or 1% air saturation, whichever is greater; 200 to 500%: ±15% of reading, relative to calibration gases
ROX™ Optical Dissolved Oxygen* mg/L	0 to 50 mg/L	0.01 mg/L	0 to 20 mg/L: ±0.1 mg/L or 1% of reading, whichever is greater; 20 to 50 mg/L: ±15% of reading, relative to calibration gases
Dissolved Oxygen** % Saturation ET✓	0 to 500%	0.1%	0 to 200%: ±2% of reading or 2% air saturation, whichever is greater; 200 to 500%: ±6% of reading
Dissolved Oxygen** mg/L ET✓	0 to 50 mg/L	0.01 mg/L	0 to 20 mg/L: ±0.2 mg/L or 2% of reading, whichever is greater; 20 to 50 mg/L: ±6% of reading
Conductivity*** 6560 Sensor* ET✓	0 to 100 mS/cm	0.001 to 0.1 mS/cm (range dependent)	±0.5% of reading + 0.001 mS/cm
Salinity	0 to 70 ppt	0.01 ppt	±1% of reading or 0.1 ppt, whichever is greater
Temperature 6560 Sensor* ET✓	-5 to +50°C	0.01°C	±0.15°C
pH 6561 Sensor* ET✓	0 to 14 units	0.01 unit	±0.2 unit
ORP	-999 to +999 mV	0.1 mV	±20 mV
Depth Medium Shallow Vented Level	0 to 200 ft, 61 m 0 to 30 ft, 9.1 m 0 to 30 ft, 9.1 m	0.001 ft, 0.001 m 0.001 ft, 0.001 m 0.001 ft, 0.001 m	±0.4 ft, ±0.12 m ±0.06 ft, ±0.02 m ±0.01 ft, 0.003 m
Turbidity* 6136 Sensor* ET✓	0 to 1,000 NTU	0.1 NTU	±2% of reading or 0.3 NTU, whichever is greater**
Nitrate/nitrogen***	0 to 200 mg/L-N	0.001 to 1 mg/L-N (range dependent)	±10% of reading or 2 mg/L, whichever is greater
Ammonium/ammonia/ nitrogen****	0 to 200 mg/L-N	0.001 to 1 mg/L-N (range dependent)	±10% of reading or 2 mg/L, whichever is greater
Chloride****	0 to 1000 mg/L	0.001 to 1 mg/L (range dependent)	±15% of reading or 5 mg/L, whichever is greater
Rhodamine*	0-200 µg/L	0.1 µg/L	±5% reading or 1 µg/L, whichever is greater

• Maximum depth rating for all standard optical sensors is 200 feet, 61 m.
•• Rapid Pulse is only available on the 6820/6920 V2-1 (one optical port version)
••• Report outputs of specific conductance (conductivity corrected to 25° C), resistivity, and total dissolved solids are
also provided. These values are automatically calculated from conductivity according to algorithms found in *Standard
Methods for the Examination of Water and Wastewater* (ed 1989).
•••• Freshwater only, Maximum depth rating of 50 feet, 15.2 m. 6820/6920 V2-1 have 3 ISE ports, 6820/6920 V2-2 have
1 ISE port.

**In YSI AMCO-AEPA Polymer Standards.

	Range	Detection Limit	Resolution	Linearity
BGA - Phycocyanin*	~0 to 280,000 cells/mL† 0 to 100 RFU	~220 cells/mL§	1 cell/mL 0.1 RFU	R² > 0.9999**
BGA - Phycoerythrin*	~0 to 200,000 cells/mL† 0 to 100 RFU	~450 cells/mL§§	1 cell/mL 0.1 RFU	R² > 0.9999***
Chlorophyll* 6025 Sensor* ET✓	~0 to 400 µg/L 0 to 100 RFU	~0.1 µg/L§§§	0.1 µg/L Chl 0.1% RFU	R² > 0.9999****

• Maximum depth rating for all standard
optical probes is 200 feet, 61 m.
BGA = Blue-Green Algae
RFU = Relative Fluorescence Units
~ = Approximately

† Explanation of Ranges can
be found in the 'Principles of
Operation' section of the 6-Series
Manual, Rev D.

§ Estimated from cultures of *Microcystis aeruginosa*.
§§ Estimated from cultures *Synechococcus sp.*
§§§ Determined from cultures of *Isochrysis sp.* and
chlorophyll *a* concentration determined via extractions.

**Relative to serial dilution of Rhodamine WT
(0-400 µg/L).
***Relative to serial dilution of Rhodamine WT
(0-8 µg/L).
****Relative to serial dilution of Rhodamine
WT (0-500 µg/L).

YSI 6820 V2 & 6920 V2 Sonde Specifications

Medium	Fresh, sea or polluted water	Software	EcoWatch*
Temperature	Operating Storage	Dimensions 6820 V2 6920 V2	Diameter Length Weight
Communications	RS-232, SDI-12	Power	External Internal

-5 to +50°C
-10 to +60°C

2.86 in, 7.3 cm | 2.85 in, 7.24 cm
13.5 in, 34.3 cm | 18 in, 45.7 cm
3.4 lbs, 1.5 kg | 4 lbs, 1.8 kg

12 V DC
8 AA-size alkaline batteries